FlashKDA SM100

B300

01

02

03

HMMA tcgen05

02

C1

FlashKDA **SM100**

01

02

agent IR

03

B300 SM100

03

KDA

CHUNK = 16

16×16 SM80 MMA

output final state state handoff

chunk K2 head value slice sequence

HMMA

FlashKDA — **CUTLASS** —

sm_103a — .so

01

02 SHA-256

03 Python

04 CUDA

05

3,640

HMMA

0

SM100 tcgen05 UTCMMA

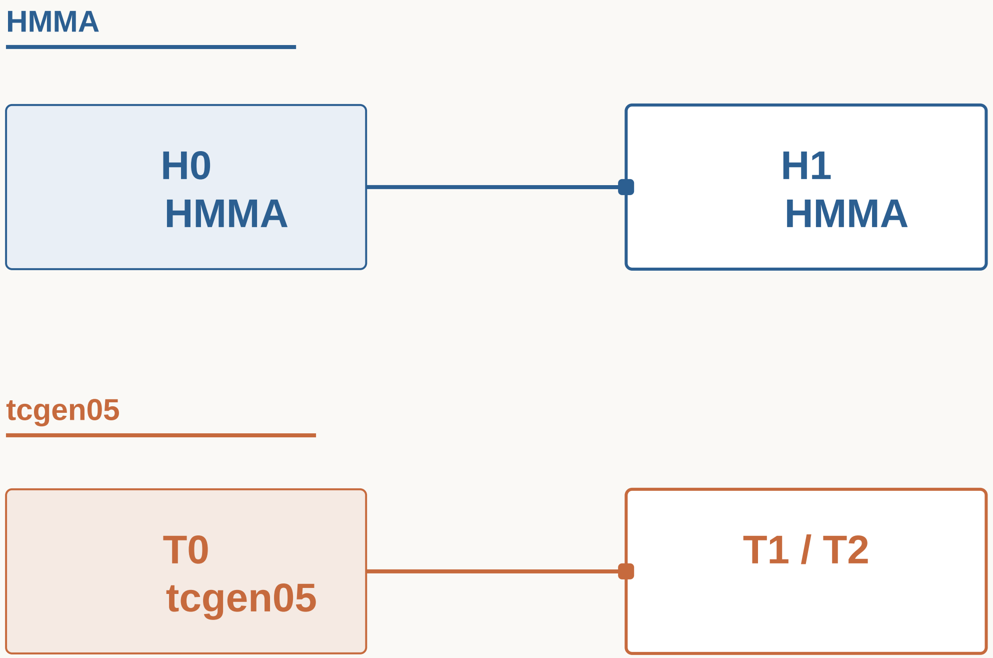
05

01

02 CUDA

03

04 L2 CUDA CUPTI ABBA



q / k / v /   /

MARPE

MARPE

Multi-Agent Runtime Profile Evolution

01

02

HMMA
tcgen05

03

GPU

04

ABBA

05

IR IR

. . . .

PIKE Atrex ABBA IR

IR

Program IR

01

02

03

04

05 Python

Experience IR

01

02

03

04

05

1 V16 1.615×
2 HMMA nseq3 1.076× nseq4 0.962×

Q1 · Q5

CHUNK16 16×16 BF16 C32/C64

Q2 · Q4

tcgen05 TMEM H12

V128 0.920× /
2.64%/1.24%

Q3 · Q6

/

H12 value − 27.0%

01

Direct

V128

0.920×

02

V16

1.615× / 0.779×

03

H3

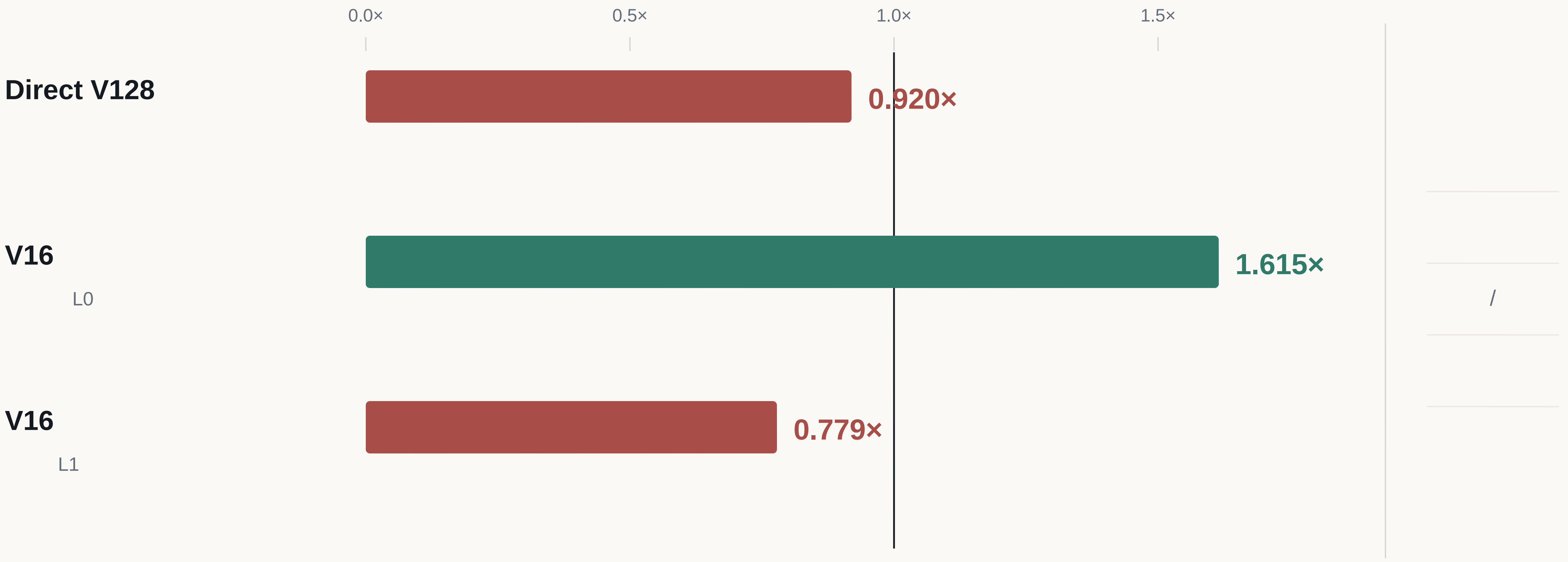
H12 0.906×

04

CAKE

H12 2.482×

tcgen05

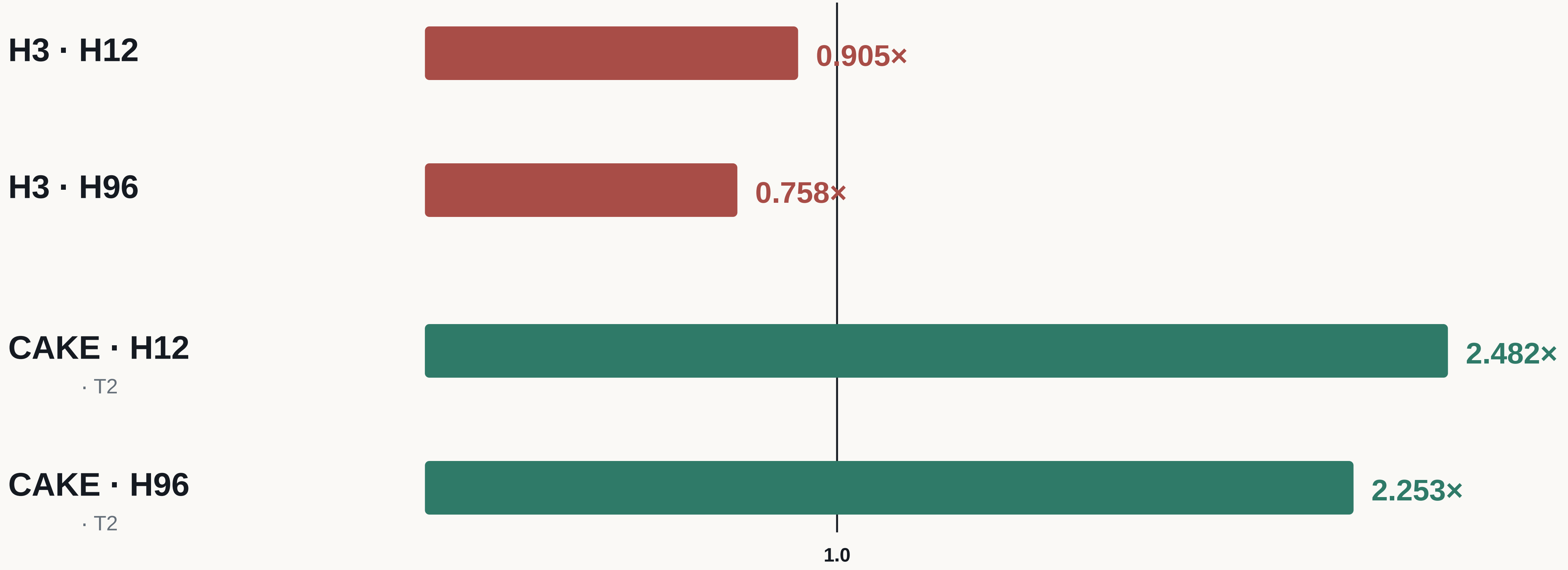


tcgen05

CAKE

= /

H3 CAKE



HMMA

nseq3 V32/V64	**1.076×**
skew3 V32/V64	**1.077×**
nseq4 V32/V64	**0.962×**

tcgen05

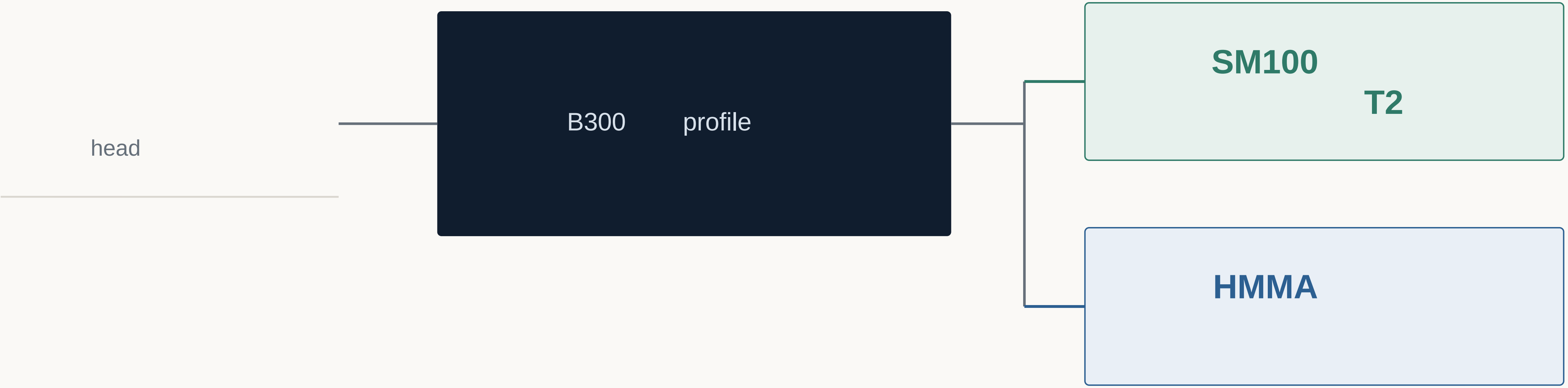
H96 s173 CAKE	**1.033×**
	0.938×
/	**1.227×**
/ HMMA	**0.955×**

IR

01 **HMMA** nseq3 7.6%

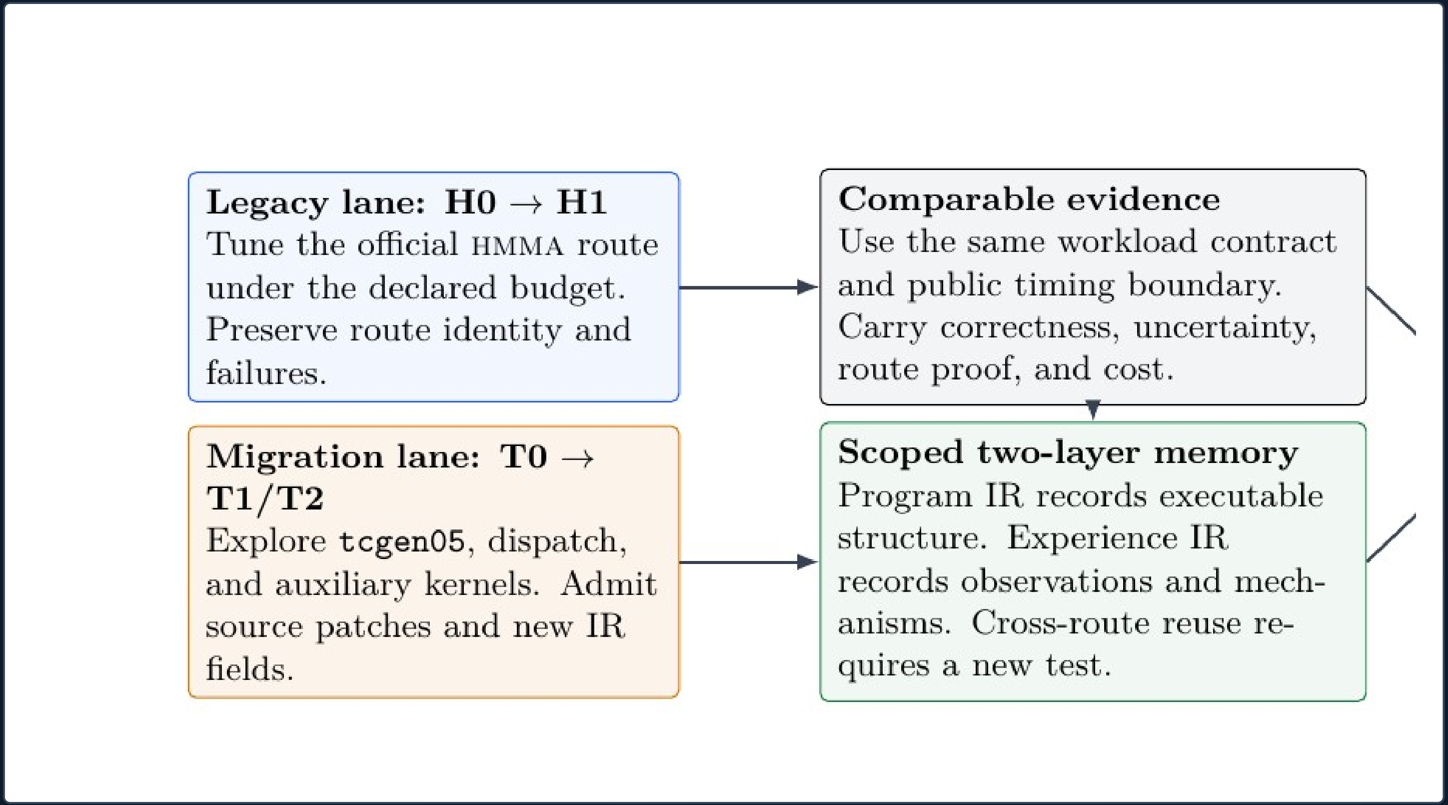
02 V32/V64 nseq3 nseq4

03 HMMA 0.955×



B300 SM100 T2

HMMA



HMMA tcgen05

SM100 profile HMMA

SM100 **FlashKDA**